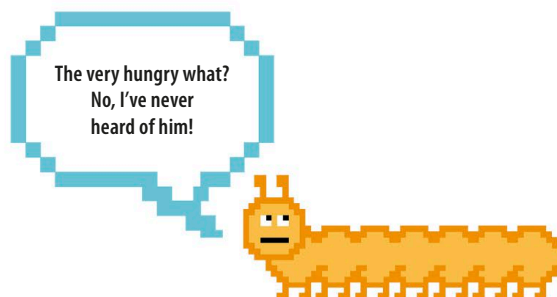


8 Get moving

The main loop moves the caterpillar forwards slightly, before performing two checks. It first checks if the caterpillar has reached the leaf. If the leaf has been eaten, the score increases, a new leaf gets drawn, and the caterpillar gets longer and faster. The loop then checks if the caterpillar has left the window – if so, the game's over. Add the main loop below the code you typed in Step 7.



```
place_leaf()
```

```
while True:
```

```
    caterpillar.forward(caterpillar_speed)
```

```
    if caterpillar.distance(leaf) < 20:
```

```
        place_leaf()
```

```
        caterpillar_length = caterpillar_length + 1
```

```
        caterpillar.shapesize(1, caterpillar_length, 1)
```

```
        caterpillar_speed = caterpillar_speed + 1
```

```
        score = score + 10
```

```
        display_score(score)
```

```
    if outside_window():
```

```
        game_over()
```

```
        break
```

The caterpillar eats the leaf when it's less than 20 pixels away.

The current leaf has been eaten, so add a new leaf.

This will make the caterpillar grow longer.

9 Bind and listen

Now put these lines below the function you've just created. The `onkey()` function binds the space-bar to `start_game()`, so you can delay the start until the player presses space. The `listen()` function allows the program to receive signals from the keyboard.

```
t.onkey(start_game, 'space')
```

```
t.listen()
```

```
t.mainloop()
```

When you press the space-bar, the game begins.

10 Test your code

Run the program. If your code is correct, you should see the caterpillar moving after you press the space-bar. Eventually, it should crawl off the screen. If the program doesn't work, check your code carefully for bugs.

